

Another reason why someone may pay a rebate to post is that rather than have to increment the bid price to be first in the queue (at a 1 cent increment), the investor may decide to enter the order to an exchange where they pay to post (inverted pricing model) to almost ensure themselves top of the queue because the counterparty would rather transact with their order and receive a rebate than transact with another order on a different venue and pay a fee. Here, the rebate cost of posting the order will be lower than the 1 cent that they would need to increment the bid even after accounting for any rebate they may receive.

In times of the NYSE and NASDAQ (well really ARCA and INET for algorithms) order routing was a much easier problem. But now with multiple destinations, venues, dark pools, etc., this is a much more complex problem.

In addition to maintaining order routing data across the various exchanges and computing the probabilities of executions, the SOR is also responsible for all order submission rules. The more common pricing rules as mentioned are market, limit, marketable limit, and floating prices that are pegged to a reference price such as the bid, ask, or midpoint and change with the reference price, etc. Varying these order types allows the algorithm to adhere to the optimally prescribed strategy by executing aggressively (i.e., market orders) and/or passively (i.e., limit orders) when needed. Order sizes are set in quantities that can be easily absorbed by the market. The order type variation disguises the actual trading intentions (i.e., limit orders) and minimizes potential adverse price impact (i.e., market orders). A reserve (iceberg) order is another technique commonly used in micro pricing strategies and refers to a continuously replenishing order at a stated size. For example, a 10,000 share order could be entered as a 1000 share reserve book order where 1000 shares would be displayed and immediately replenished each time it transacts until the order is filled. Finally, the SOR maintains, randomizes, and varies wait and cancellation times to help disguise trading intentions. Some orders may remain in the market for longer periods of time while other orders remain on the book for a shorter period of time before cancellation. Additionally, randomizing the time between orders and waves helps hide trading intentions, minimizes information leakage, and helps improve the likelihood of achieving more favorable prices.

An important note that is often overlooked in the algorithmic trading arena is that a smart order router should only be used as a smart order router. Many vendors have made the mistake of forcing the smart order router to provide a combination of services such as limit order model,